AMCS 206

Problem Set #5

Problem 1

$$\begin{cases} 0.003000\, x_1 + 59.14 x_2 = 59.17 & (1) \\ 5.291\, x_1 - 6.130\, x_2 = 46.78 & (2) \end{cases}$$

1st step:

- 1st pivot is $a_{11}^{(1)} = 0.003000$
- associated multiplier is $m_{21} = \dfrac{5.291}{0.003000} = 1763.67$

- $(2) - m_{21}(1) = (2)$ yields (without rounding):

$$\boxed{\begin{aligned} 0.003000\, x_1 + 59.14 x_2 &= 59.17 \\ -104309.3767 x_2 &= -104309.3767 \end{aligned}} \quad (3)$$

using 4-digits arithmetic with rounding:

$$fl(1763.67) = 0.1763 \times 10^4$$
$$fl(-6.130) = -0.6130 \times 10^1$$
$$fl(59.14) = 0.5914 \times 10^2$$

$$\begin{aligned} & fl(fl(-6.130) - fl(1763.67) \times fl(59.14)) \\ = \; & fl(-0.6130 \times 10^1 - 0.1763 \times 10^4 \times 0.5914 \times 10^2) \\ = \; & fl(-0.6130 \times 10^1 - 0.1043 \times 10^6) \\ = \; & fl(-0.6130 \times 10^1 \times 10^5 \times 10^{-5} - 0.1043 \times 10^6) \\ = \; & fl(-0.0000 \times 10^6 - 0.1043 \times 10^6) \\ = \; & -104300 \end{aligned}$$

in the same way:

$$fl(fl(46.78) - fl(1763.67) \times fl(59.17)) = -104400$$

so we have

$$\boxed{\begin{aligned} 0.003000\, x_1 + 59.14 x_2 &= 59.17 \\ -104300\, x_2 &= -104400 \end{aligned}} \quad (4)$$

backward substitution (the round-off error propagates):

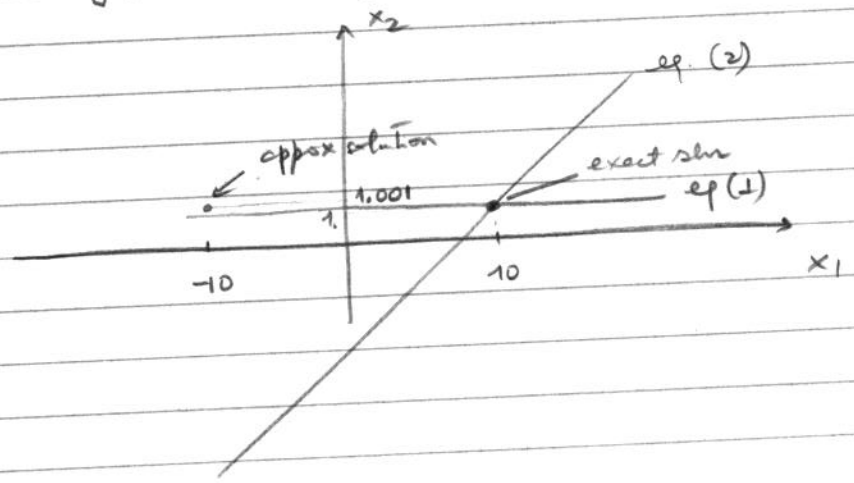
$$x_2 = fl(-0.1044 \times 10^6 \;/\; -0.1043 \times 10^6)$$
$$= fl(0.1001 \times 10^1)$$
$$= 1.001$$

$x_1 \simeq -10.0 \neq 10.0$ which is the exact value

What happened? notice that the small error 0.001 was multiplied by a large number:

$$\frac{59.14}{0.003000} \simeq 20000.$$

graphically, it is easy to show what happens for a small syst:



Problem 2

1. $$AB = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 0 \\ -1 & 1 & 2 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

$$= \begin{bmatrix} b_{11} + 2b_{21} - b_{31} & b_{12} + 2b_{22} - b_{32} & b_{13} + 2b_{23} - b_{33} \\ 2b_{11} + b_{21} & 2b_{12} + b_{22} & 2b_{13} + b_{23} \\ -b_{11} + b_{21} + 2b_{31} & -b_{12} + b_{22} + 2b_{32} & -b_{13} + b_{23} + 2b_{33} \end{bmatrix}$$

Because $AB = I$, the 9 equations are:

$$\begin{cases} b_{11} + 2b_{21} - b_{31} = 1 \\ 2b_{11} + b_{21} = 0 \\ -b_{11} + b_{21} + 2b_{31} = 0 \end{cases} \iff A \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} b_{12} + 2b_{22} - b_{32} = 0 \\ 2b_{12} + b_{22} = 1 \\ -b_{12} + b_{22} + 2b_{32} = 0 \end{cases} \iff A \begin{bmatrix} b_{12} \\ b_{22} \\ b_{23} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{cases} b_{13} + 2b_{23} - b_{33} = 0 \\ 2b_{13} + b_{23} = 0 \\ -b_{13} + b_{23} + 2b_{33} = 1 \end{cases} \iff A \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

2. $$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

apply naive forward elimination to get

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -3 & 2 & -2 & 1 & 0 \\ 0 & 0 & 3 & -1 & 1 & 1 \end{array}\right]$$

3. Backward substitution is performed on:

(i) $$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} \Rightarrow \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} -2/9 \\ 4/9 \\ -1/3 \end{bmatrix}$$

(ii) $$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \begin{bmatrix} 5/9 \\ -1/9 \\ 1/3 \end{bmatrix}$$

i) $$\begin{bmatrix} 1 & 2 & -1 \\ 0 & -3 & 2 \\ 0 & 0 & 3 \end{bmatrix} \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} -1/9 \\ 2/9 \\ 1/3 \end{bmatrix}$$

We can use naive Gaussian elimination on $[A \mid I]$ to determine the inverse of A.

1st) Find LU factorization of A : $A = LU$

2nd) define

$$b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} ; \quad b_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} , \quad b_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

3rd) solve, for $i = 1, 2, 3$,

- $L z_i = b_i$, followed by
- $U x_i = z_i$

then $B = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix}$.

<u>Problem 3</u>

The solution is $x_1 = -0.163$
$x_2 = 5.98$
$x_3 = 0.142$.

Please check Matlab Central for codes implementing GESPP

Problem 4

$$\|A\|_\infty = \max_{1 \le i \le n} \sum_{j=1}^{n} |a_{ij}| \quad , \quad A = (a_{ij})_{1 \le i,j \le n}$$

$$\|A\|_\infty = \max\{ |1| + |2| \, , \, |1.0001| + |2| \} = 3.0001$$

$$A^{-1} = \begin{bmatrix} -10000 & 10000 \\ 5000.5 & -5000. \end{bmatrix}$$

from Matlab

$$\|A^{-1}\|_\infty = \max\{ 10000 + 10000 \, , \, 5000.5 + 5000 \} = 20000.$$

$$K(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty = 60002 \gg 1$$

so that we can say that a small residual vector <u>does not</u> imply a small error vector, or, an accurate approximate solution.